Kompass

Navigate Together





Aswin P[502319]
Jefrin J[502326]
Mohamed Feroz S [502337]
Barnabas Mathew Vaz A [502361]

Problem Statement

Tour groups currently depend on multiple applications for:

- Communication
- Navigation and location sharing
- Attendance management
- Announcements
- Emergency coordination

This results in:

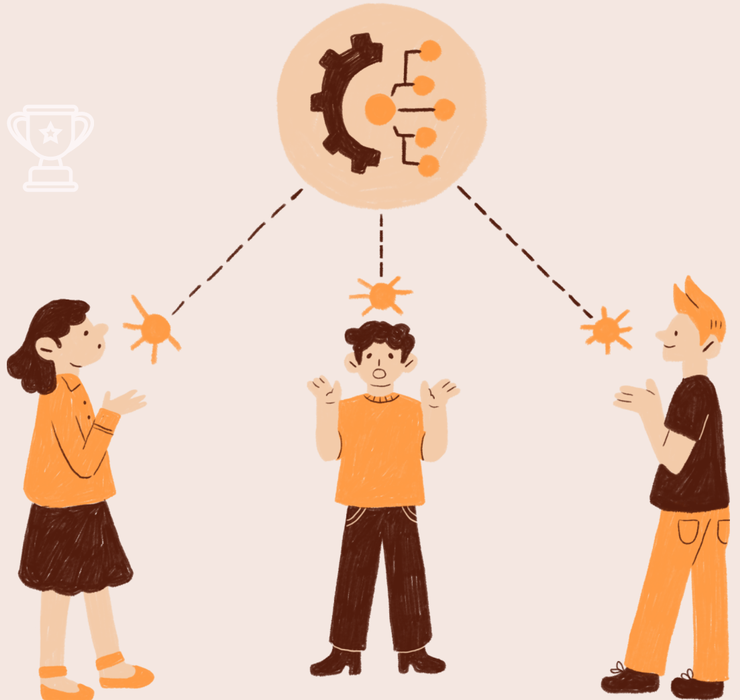
- Fragmented communication
- Difficulty identifying missing participants
- Inaccurate or time-consuming attendance
- Delayed emergency response
- Increased workload for tour organizers

Proposed Solution :

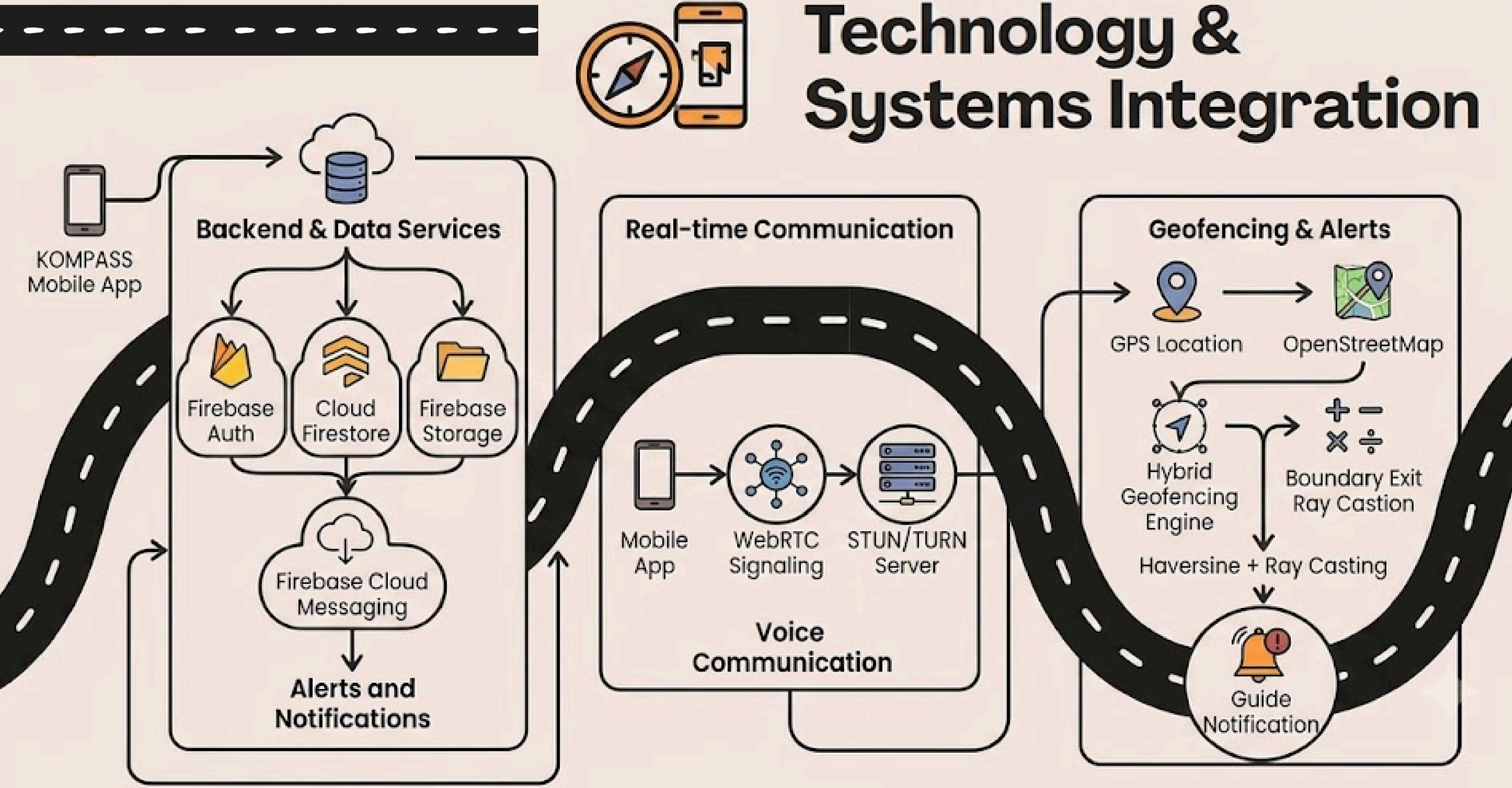


KOMPASS provides a centralized mobile platform for tour organizers, guides and participants.

- Tour and participant management
- Digital attendance using QR code, GPS or manual check-in
- Live participant location tracking
- Circular and polygonal geofencing
- In-app voice calling and messaging
- Announcements and push notifications
- Itinerary and meeting-point management
- Emergency SOS alerts
- Document and ticket sharing

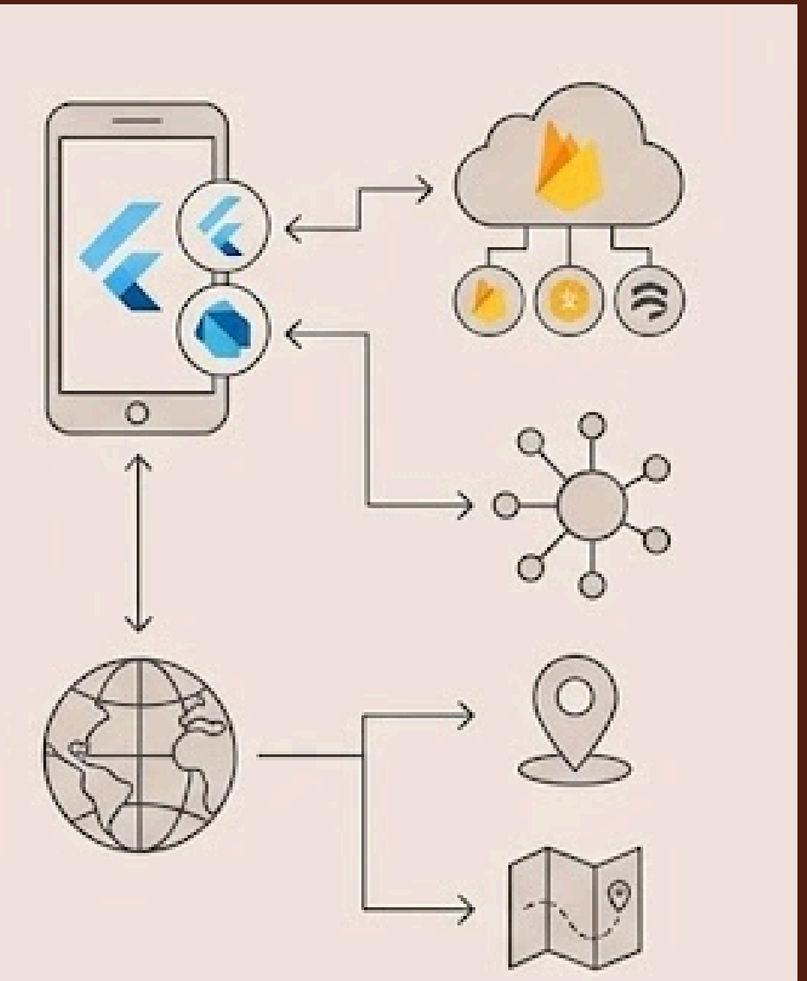


Technology & Systems Integration



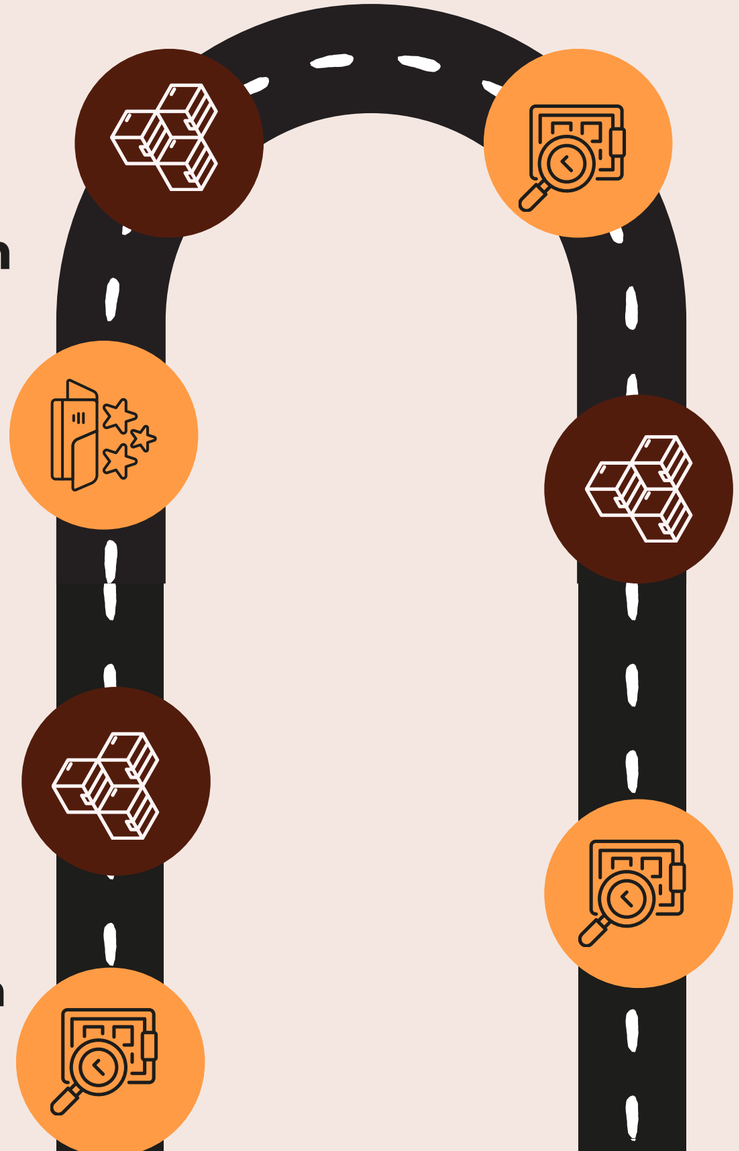
Implementation Details

Component	Technology
Mobile Application	Flutter
Programming Language	Dart
Backend and Database	Firebase and Cloud Firestore
Authentication	Firebase Authentication
Notifications	Firebase Cloud Messaging
Voice Communication	WebRTC
Map Visualization	OpenStreetMap
Location Services	Geolocator
Local Storage	Hive
State Management	Riverpod



Expected Outcomes and Future Scope

- Centralized tour coordination
- Improved emergency response
- Accurate digital attendance
- Faster communication between guides and participants



- Real-time identification of missing participants
- Safer and more organized tour experiences
- Reduced manual effort for organizers

THANK YOU